

The Levelized Cost of Energy

Comparing how much a kilowatt-hour *really* costs

PHY199-RAP · Energy in the 21st Century

What is LCOE?

The **levelized cost of energy** is the average price per unit of electricity that pays back *all* lifetime costs of a plant.

$$\text{LCOE} = \frac{\text{lifetime costs}}{\text{lifetime energy produced}}$$

It lets us compare a solar farm, a gas plant, and a reactor on the **same footing** — dollars per megawatt-hour.

A simple calculation

A solar farm costs **\$1,000,000** to build, runs **25 years**, and makes **2,500 MWh per year**.

$$\text{LCOE} = \frac{\$1,000,000}{25 \times 2,500 \text{ MWh}} = \$16 / \text{MWh}$$

(Real estimates add fuel, maintenance, and a discount rate — but the idea is the same.)

Rough LCOE by source

Source	Typical LCOE (\$/MWh)
Utility solar	30 – 40
Onshore wind	25 – 50
Natural gas	45 – 75
Nuclear	55 – 90
Coal	65 – 150

Cheapest on paper isn't the whole story — **land, waste, and human cost** don't appear here.

Takeaway

LCOE is a powerful *first* comparison —
but a fair energy choice weighs the costs it **leaves out**.